Byeonghun Lee

PhD student
Korea University, School of Electrical Engineering
145 Anam-ro, Seongbuk-gu, Seoul, 02841, Republic of Korea
byeonghun_lee@korea.ac.kr
+82) 10-4608-7133

EDUCATION

Mar. 2017 ~ Feb. 2022	**Daegu Gyeongbuk Institute of Science and Technology** School of Undergraduate Studies *B.S. in Convergence* GPA: 3.21 / 4.3	Daegu, Korea
Mar. 2022 ~ Present	**Korea University** School of Electrical Engineering *Advisor: Kyong Hwan Jin* *Ph.D. Student* GPA: 4.38 / 4.5	Seoul, Korea

RESEARCH INTERESTS

- Image processing
- Inverse problem
- Retrieval-Augmented Generation in Vision
- Image Super-Resolution
- 3D rendering, Depth Estimation

PUBLICATIONS

1. **Byeonghun Lee**, Jaedeok Kim, Sunghoon Im and Kyong Hwan Jin, "Target-Oriented Neural MPI Stitching for Novel View Synthesis," IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2027.
2. **Byeonghun Lee**, Hyunmin Cho, Sunghoon Im†, and K.H. Jin†, "ReAL: Reference-to-Image (R2I) Aware Latent Diffusion for Image Super-Resolution," European Conference on Computer Vision (ECCV), 2026.
3. **Byeonghun Lee***, Hyunmin Cho*, Honggyu Choi, Soo Min Kang, Iljun Ahn and K.H. Jin*, "Reference-based Super-Resolution via Image-based Retrieval-Augmented Generation Diffusion," IEEE/CVF International Conference on Computer Vision (ICCV), 2025.
4. Woo Kyoung Han*, Yongjun Lee*, **Byeonghun Lee**, Sang Hyun Park, Sunghoon Im*, and K.H. Jin*, "JPEG Processing Neural Operator for Backward-Compatible Coding," IEEE/CVF International Conference on Computer Vision (ICCV), 2025.

5. Woo Kyoung Han, **Byeonghun Lee**, Hyunmin Cho, Sunghoon Im*, K.H. Jin*, "Towards Lossless Implicit Neural Representation via Bit Plane Decomposition," IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
6. EungGu Kang, **Byeonghun Lee**, Sunghoon Im*, and K. H. Jin*, "BurstM: Deep Burst Multi-scale SR using Fourier Space with Optical Flow," European Conference on Computer Vision (ECCV), 2024.
7. Minsu Kim, Jaewon Lee, **Byeonghun Lee**, Sunghoon Im and K. H. Jin, "Implicit Neural Image Stitching With Enhanced and Blended Feature Reconstruction," IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2024.
8. Woo Kyoung Han, **Byeonghun Lee**, Sang Hyun Park, and K. H. Jin, "ABCD : Arbitrary Bitwise Coefficient for De-quantization," IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023.

PATENTS

1. Apparatus and Method of Recovering Image Using Arbitrary Bitwise Coefficient Estimation for De-quantization, 1020230111027(2023-08-24)
2. Method and Apparatus for Lossless Implicit Neural Representation using Bit-plane Decomposition, 1020250151831(2025-10-20)
3. Polarization-related patents (**Samsung Electronics collaboration**)
   - 1 international patent application, pending

PROJECTS

- Synthetic Polarization Data Generation from RGB Images & Polarization-Based Precise Depth Estimation — **Samsung Electronics, GTR** (MAR 2026 –)
- Hallucination Mitigation in Video Understanding Models — **Samsung Research** (DEC 2025 –)
- User-Preference-Based Image Quality Enhancement — **Samsung Research** (Jun 2024 – May 2025)
- Cognitive Reverse-Haptics Reproduction System Lab — **Basic Research Laboratory (BRL), National Research Foundation of Korea** (Feb 2023 – Feb 2024) Developed signal-processing algorithms to convert real-world tactile data into digital signals, and designed a Verilog-based hardware demonstration.
- Food Classification Algorithm for Caloric Intake Analysis — **Pinset Healthcare** (Aug 2022 – Jul 2023) Developed food classification and calorie estimation algorithms from user-captured food images.
- Deep Learning Algorithm for Bacteria Detection — **THE WAVE TALK** (Mar 2022 – Aug 2022) Developed an image-based classification algorithm to identify bacteria species and concentration levels.

AWARDS AND HONORS

- Bronze medal for paper award , IPIU, Korea (Jun. 2026)

SKILLS AND TECHNIQUES

- Proficient in C++ and Python
- Experienced in developing AI algorithms with PyTorch
- Background in 3D rendering and image warping
- Expertise in image super-resolution techniques

- Research and development experience with diffusion models